

Self-Built Simple Quadruped Robot Dog - Final Presentation

This English version is formatted for graduate application review. It preserves the engineering purpose, design decisions, validation evidence, and individual contribution signals of the original course document while presenting the material in concise English.

Applicant	CHENG-HSUAN LEE
Project	Self-Built Simple Quadruped Robot Dog
Course	Microprocessor Controlled Systems
Document Type	Final presentation deck
Primary Role	Team lead, presenter, and integration contributor
Original Source	robot-dog-final-presentation.pdf (33 pages)

Executive Summary

This English presentation version summarizes the final robot dog build, including structure, electronics, gait logic, mechanical updates, and behavior demonstrations.

The presentation evidence shows a complete small-scale legged robot workflow: design, fabricate, wire, simulate, program, calibrate, repair, and demonstrate.

Applicant Contribution Signals

- Explained the project as an integrated robot system, not separate slides of CAD and code.
- Connected behavior demonstrations to the mechanical and embedded decisions behind them.
- Presented technical progress in a form that reviewers can scan quickly.

Presentation Narrative

The project begins with the requirement to create an interactive quadruped robot and moves through body design, servo selection, control logic, and testing.

Final behaviors demonstrate that the platform could execute more than one scripted motion after calibration.

Reviewer Focus

Look for cross-domain execution: 3D-printed structure, electronics layout, MATLAB reasoning, ESP32/PCA9685 control, and repeated hardware debugging.

Technical Decision Table

Constraint	Decision	Why It Mattered	Skill Demonstrated
Audience needs	Condensed technical workflow into presentation form	Made engineering decisions easy to review	Technical communication
Demonstration focus	Linked behaviors to underlying control	Showed system-level execution	Robotics integration
Final result	Presented multiple working behaviors	Supported A+ course outcome	Execution quality

How to Read This Evidence

This document is intended to help a reviewer quickly understand the engineering content of the original report in English. The original PDF remains available in the project archive as the source artifact with its original figures, tables, screenshots, and course formatting.